Experiment No 6

Windows Operating System Commands Execution

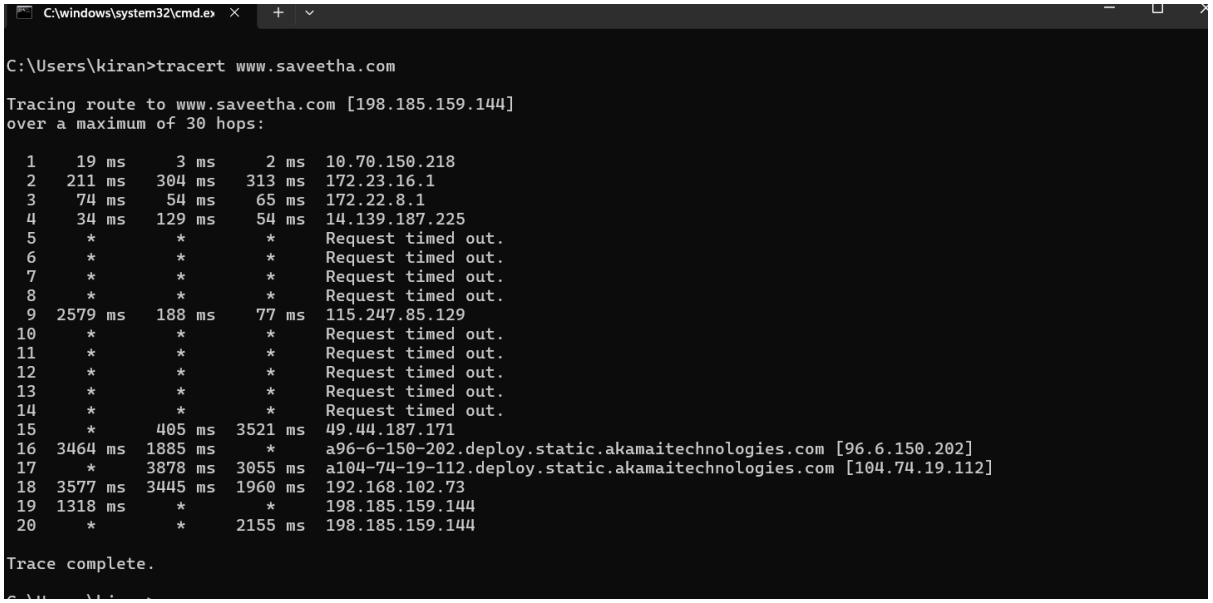
Aim

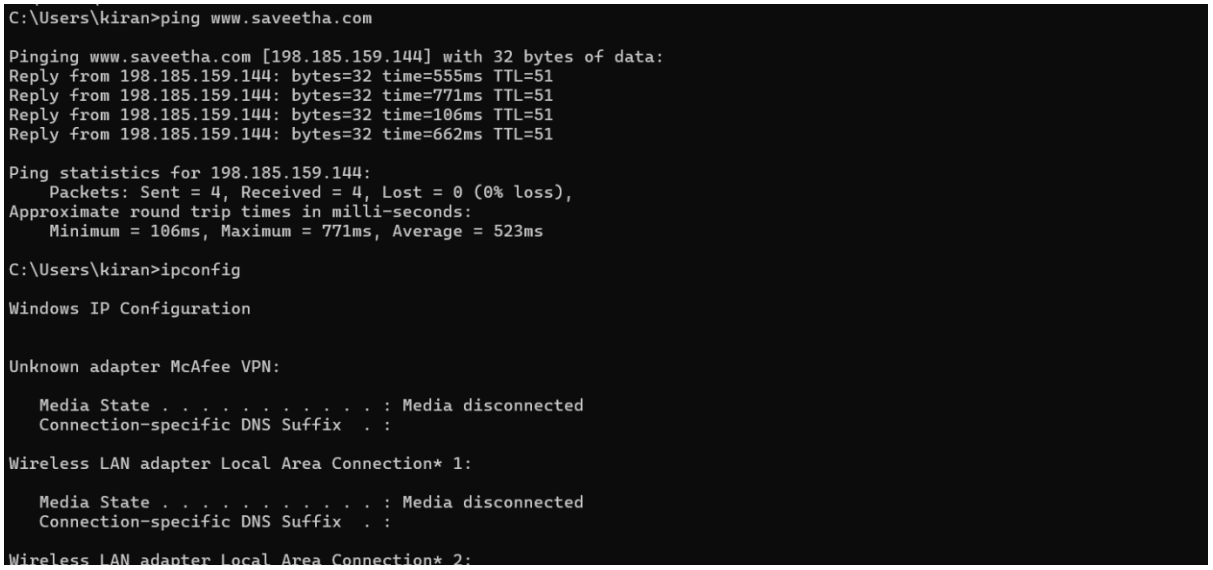
To execute networking commands such as traceroute, ping, ifconfig, and netstat in Kali Linux to analyze network connectivity, routing, IP configuration, and active network connections.

Algorithm

1. Start the Kali Linux operating system and open the Terminal.
2. Execute the traceroute www.saveetha.com command to identify the route taken by packets to reach the destination.
3. Execute the ping -c 4 www.saveetha.com command to check network connectivity.
4. Execute the ifconfig command to display network interface configuration and IP addresses.
5. Execute the netstat command to display network connections and related information.
6. Observe and record the output of each command.

Output


C:\windows\system32\cmd.ex
C:\Users\kiran>tracert www.saveetha.com
Tracing route to www.saveetha.com [198.185.159.144]
over a maximum of 30 hops:
1 19 ms 3 ms 2 ms 10.70.150.218
2 211 ms 304 ms 313 ms 172.23.16.1
3 74 ms 54 ms 65 ms 172.22.8.1
4 34 ms 129 ms 54 ms 14.139.187.225
5 * * * Request timed out.
6 * * * Request timed out.
7 * * * Request timed out.
8 * * * Request timed out.
9 2579 ms 188 ms 77 ms 115.247.85.129
10 * * * Request timed out.
11 * * * Request timed out.
12 * * * Request timed out.
13 * * * Request timed out.
14 * * * Request timed out.
15 * 405 ms 3521 ms 49.44.187.171
16 3464 ms 1885 ms * a96-6-150-202.deploy.static.akamaitechnologies.com [96.6.150.202]
17 * 3878 ms 3055 ms a104-74-19-112.deploy.static.akamaitechnologies.com [104.74.19.112]
18 3577 ms 3445 ms 1960 ms 192.168.102.73
19 1318 ms * * 198.185.159.144
20 * * 2155 ms 198.185.159.144
Trace complete.


C:\Users\kiran>ping www.saveetha.com
Pinging www.saveetha.com [198.185.159.144] with 32 bytes of data:
Reply from 198.185.159.144: bytes=32 time=555ms TTL=51
Reply from 198.185.159.144: bytes=32 time=771ms TTL=51
Reply from 198.185.159.144: bytes=32 time=106ms TTL=51
Reply from 198.185.159.144: bytes=32 time=662ms TTL=51
Ping statistics for 198.185.159.144:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 106ms, Maximum = 771ms, Average = 523ms
C:\Users\kiran>ipconfig
Windows IP Configuration
Unknown adapter McAfee VPN:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 2:

```
Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::751d:aab4:ac44:ed58%7
   IPv4 Address. . . . . . . . . . . : 10.70.150.63
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.70.150.218

C:\Users\kiran>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.70.150.63:49377     ec2-3-6-199-65:https   FIN_WAIT_1
  TCP    10.70.150.63:49700     4.213.25.241:https     ESTABLISHED
  TCP    10.70.150.63:49701     4.213.25.240:https     ESTABLISHED
  TCP    10.70.150.63:49932     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:50060     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:50752     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:50944     172.188.155.25:https   ESTABLISHED
  TCP    10.70.150.63:51088     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:51487     172.64.148.235:https   ESTABLISHED
  TCP    10.70.150.63:51997     4.213.25.241:https     ESTABLISHED
  TCP    10.70.150.63:52090     172.188.155.25:https   ESTABLISHED
  TCP    10.70.150.63:52326     13.67.10.229:https     ESTABLISHED
  TCP    10.70.150.63:51997     4.213.25.241:https     ESTABLISHED
  TCP    10.70.150.63:52090     172.188.155.25:https   ESTABLISHED
  TCP    10.70.150.63:52326     13.67.10.229:https     ESTABLISHED
  TCP    10.70.150.63:53731     104.18.32.47:https     ESTABLISHED
  TCP    10.70.150.63:54433     whatsapp-cdn-shv-02-maa5:https  ESTABLISHED
  TCP    10.70.150.63:54594     cn-in-f95:https        ESTABLISHED
  TCP    10.70.150.63:54852     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:55663     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:56703     lcbomp-in-f84:https    ESTABLISHED
  TCP    10.70.150.63:57896     pnmaaa-am-in-f14:https  ESTABLISHED
  TCP    10.70.150.63:59441     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:59511     maa05s16-in-f14:https  TIME_WAIT
  TCP    10.70.150.63:59657     a23-220-237-244:https  CLOSE_WAIT
  TCP    10.70.150.63:59751     lcbomp-in-f84:https    TIME_WAIT
  TCP    10.70.150.63:59918     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:60065     1:https                ESTABLISHED
  TCP    10.70.150.63:61407     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:61422     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:61855     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:62452     104.18.39.21:https     ESTABLISHED
  TCP    10.70.150.63:62472     pnmaaa-an-in-f14:https  TIME_WAIT
  TCP    10.70.150.63:62824     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:64849     sm-in-f188:5228        ESTABLISHED
  TCP    10.70.150.63:64918     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:65044     lb-140-82-114-25-iad:https  TIME_WAIT
  TCP    10.70.150.63:65228     10.70.150.218:domain   TIME_WAIT
  TCP    10.70.150.63:65351     104.18.39.21:https     ESTABLISHED
  TCP    10.70.150.63:65373     cn-in-f95:https        ESTABLISHED
```

Result

The networking commands traceroute, ping, ifconfig, and netstat were executed in Kali Linux to analyze network routing, connectivity, IP configuration, and network connections.